

Name: _____

Practice Quiz for CS599 B1: Advanced Natural Language Processing

Instructions:

- You will have 20 minutes at the start of class to complete the quiz.
- This quiz is to be completed individually by each student.
- You are **not** allowed any reference material, calculators, or other electronic devices.
- Answer these questions by circling the correct option or writing the answer. **If given letter options, give exactly one answer. Carefully read the instructions on each question.**

1. (10 points)

- _____ 1. What is the correct order of operations in the Transformer block you implemented? (Note: this is *not* the one you implemented in your ablation experiments.)
- A. RMSNorm, SwiGLU MLP, RMSNorm, Self-attention
 - B. Self-attention, RMSNorm, SwiGLU MLP, RMSNorm
 - C. SwiGLU MLP, RMSNorm, Self-attention, RMSNorm
 - D. RMSNorm, Self-attention, RMSNorm, SwiGLU MLP
- _____ 2. Let d_k , d_v , and d_{model} denote vector dimensions per head, and let h denote the number of heads. How many parameters are in the multi-head self attention layer? Do not make any assumptions about the relation between d_k and d_{model} . Ignore bias terms in linear layers.
- _____ 3. Which of the following is the most complete and accurate description of what a SwiGLU contains? Select only one.
- A. A sigmoid activation function
 - B. A Swish activation function and linear gate
 - C. A linear gate
 - D. A Swish activation function, linear gate, and ReLU activation function
- _____ 4. Given the string “l a t e _ l a t s _ l a m p” that has been split into characters, what would be the first **two** tokens added to the vocabulary when running BPE? Assume that you cannot merge “_” characters, nor merge across them.
- _____ 5. When using rotary position embeddings (RoPE), which of the following is true?
- A. The key at position 5 and query at position 10 will have the same positional relationship as the key at position 15 and query at position 20.
 - B. The token at position 100 will have the same RoPE encoding as the token at position 1000.
 - C. RoPE is added to the input embeddings before the first Transformer block.
 - D. At a given position, every dimension in the query vector is rotated to the same degree.

- _____ 6. Given the vocabulary [a, b, c, d, e] with probability distribution [0.1, 0.4, 0.3, 0.05, 0.15], which tokens could you sample if you were to use nucleus sampling with $p = 0.9$?
- _____ 7. In 1–2 sentences, give one concrete reason why we might want to train a BPE tokenizer on UTF-8 bytes instead of characters.
- _____ 8. As a function of the number of parameters P in your language model, how does the memory usage of AdamW scale?
- A. $O(1)$
 - B. $O(\sqrt{P})$
 - C. $O(P)$
 - D. $O(P^2)$
- _____ 9. For which of the following operations is it most important to use `float32` precision, rather than `float16`?
- A. Linear layer
 - B. The Swish function
 - C. RMSNorm
 - D. The input embedding layer
- _____ 10. What is the correct order of operations in AdamW?
- A. Compute gradient, compute and apply weight update, perform moment updates
 - B. Compute gradient, perform moment updates, compute and apply weight update
 - C. Perform moment updates, compute gradient, compute and apply weight update
 - D. Perform moment updates, compute and apply weight update, compute gradient